OVER A DOMAIN, FLATNESS OF A FINITE TYPE ALGEBRA AND OF A FINITE FLAT MODULE FORCES FINITE PRESENTATION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Flatness of a finite type algebra over an integral base forces finite presentation. Same chapter flat.tex, label lemma-finite-type-flat-over-integral (invoked at proof line 4222). It is exactly Part (1), applied to the algebra S over the domain R.
- The same statement for modules, on the level of one polynomial algebra. Same chapter flat.tex, label lemma-finite-type-flat-algebra (invoked at proof line 4224). Applied after choosing a surjection $R[x_1, \ldots, x_n] \to S$, it makes the finite module M finitely presented over the polynomial ring.
- Descent of finite presentation from a ring to a quotient. Algebra, label algebra-lemma-finitely-presented-over-subring (invoked at proof line 4227). It converts finite presentation over $R[x_1, \ldots, x_n]$ into finite presentation as an S-module, which is the conclusion of Part (2).

Source reference. Original lemma 0561: sources/stacks-chow/tags.txt, source lines 6694–6694. Navigation pointer for \ref{algebra-lemma-finitely-presented-over-subring} used inside the counted span (line 4227); the argument that consumes it is retained in the span.

Source reference. Original lemma 05IL: sources/stacks-flat/pinned-original-source.tex, source lines 4162–4162. Navigation pointer for \ref{lemma-finite-type-flat-algebra} used inside the counted span (line 4224); the argument that consumes it is retained in the span.

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Source reference. Original lemma 05IN: sources/stacks-flat/pinned-original-source.tex, source lines 4198–4198. Navigation pointer for \ref{lemma-finite-type-flat-over-integral} used inside the counted span (line 4222); the argument that consumes it is retained in the span.

Proposition 0.1. *Let R be a domain. Let $R \to S$ be a ring map of finite type. Let M be a finite S-module.*

(1) *If S is flat over R, then S is a finitely presented R-algebra.*
(2) *If M is flat as an R-module, then M is finitely presented as an S-module.*

Proof. Part (1) is a special case of Lemma 05IN. For Part (2) choose a surjection $R[x_1, \ldots, x_n] \to S$. By Lemma 05IL we find that M is finitely presented as an $R[x_1, \ldots, x_n]$-module. We conclude by Algebra, Lemma 0561. □